

WhatsApp Chat Analysis – Problem Statement

You are given raw WhatsApp chat data (exported text file) containing messages, timestamps, users, and media indicators. The objective is to analyze this unstructured chat data to extract meaningful insights about communication patterns, user activity, and engagement trends. Manual analysis of chat data is inefficient due to its unstructured nature. Therefore, an automated solution is required to process and visualize chat insights effectively.

Objective

- 1 Analyze WhatsApp chat data to understand user behavior
- 2 Generate insights like message count, word usage, and activity trends
- 3 Visualize chat patterns using interactive charts
- 4 Identify most active users and communication peaks
- 5 Provide an easy-to-use dashboard for users

Key Business Questions

- 1 What is the total number of messages, words, media, and links shared?
- 2 Who are the most active users in the chat?
- 3 How does messaging activity vary daily and monthly?
- 4 Which day and month have the highest activity?
- 5 At what time are users most active (heatmap analysis)?
- 6 What are the most commonly used words in conversations?
- 7 Which emojis are used the most?
- 8 How does user engagement change over time?

Challenges

- 1 Handling unstructured chat format
- 2 Cleaning irrelevant system messages and media placeholders
- 3 Processing multilingual text
- 4 Extracting meaningful patterns from noisy data

Expected Outcome

- 1 Interactive dashboard built using Streamlit
- 2 Clear visualizations including timelines, heatmaps, and charts
- 3 Identification of key users and peak activity times
- 4 Actionable insights into communication behavior